



National Teachers College

629 J Nepomuceno, Quiapo, Manila, 1001 Metro Manila
Bachelor of Science in Information Technology

Activity 6: Student Management Web App

Rim's Student Management System (RSMS)

A Web Development Project

*Presented to the Faculty of the
College of Information Technology*

**In Partial Fulfillment of the Requirements for the Course
Web Development**

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Rim's Student Management System

Objectives

- Create a simple multi-page web application using HTML, CSS, and JavaScript
- Design a user-friendly interface with proper layout, navigation, and styling
- Apply JavaScript fundamentals such as variables, arrays, and objects
- Use array methods including push(), shift(), splice(), includes(), and join() to manage data
- Implement the spread operator to update object properties
- Perform basic DOM manipulation to display and update content
- Allow users to add, view, search, and modify student records

Tools Used

Languages:	<u>HTML, CSS, JavaScript</u>
Online References:	<u>GeeksforGeeks, W3Schools</u>
Web Browser:	<u>Google Chrome</u>
Text Editor:	<u>VSCode, Notepad</u>

HTML Structures

1. Home Page (index.html)

This page serves as the homepage of the Student Management System. It introduces the system and provides navigation to other pages.

- For this page, I implemented:
- Header for system title
- Navigation menu/links,
- Animated GIF for visual design,
- Welcome message and system description
- Footer.

Whole Code Snippet:

```
<!DOCTYPE html>
<html>
<head>
  <title>Rim's SMS</title>
</head>

<body>
```

```

<header>
  <h1><strong>Rim's Student Management System</strong></h1>
  <hr>
  <nav>
    <a href="index.html">Home</a>
    <a href="students.html">List/Edit</a>
    <a href="form.html">Add</a>
  </nav>
</header>

<div class="main-content">
  <section class="container">
    <div class="top-row">
      
      <h2>Hello, Welcome to Rim's Student Management System!</h2>
    </div>

    <p class="home-text">
      This system allows you to manage student records easily.<br>
      You can add new students, view existing records, search for specific students,
      and edit or update their information.
    </p>
  </section>
</div>

<footer>
  <p>© 2026 Rim's SMS Web | All Rights Reserved.</p>
</footer>

</body>
</html>

```

Screenshot of the web page:



Rim's Student Management System

[Home](#) [List/Edit](#) [Add](#)



Hello, Welcome to Rim's Student Management System!

This system allows you to manage student records easily.
You can add new students, view existing records, search for specific students, and edit or update their information.

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2. Student List/Edit Page (students.html)

This page displays the list of students using JavaScript. It will allow users to interact with the student data through different buttons.

Code Snippet:

```
<div class="main-content">
  <section class="container">

    <h3>List of Students</h3>
    <div id="output"></div>

    <button onclick="modifyStudent()">Modify First Student</button>
    <button onclick="addSample()">Add Sample</button>
    <button onclick="removeFirst()">Remove First</button>
    <button onclick="searchStudent()">Search Ana</button>

  </section>
</div>
```

Screenshot of the web page:



Student List

[Home](#) [List/Edit](#) [Add](#)

List of Students

Modify First Student Add Sample Remove First

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3. Add Student Form Page (form.html)

This page will allow users to add new student data into the system.

Code Snippet:

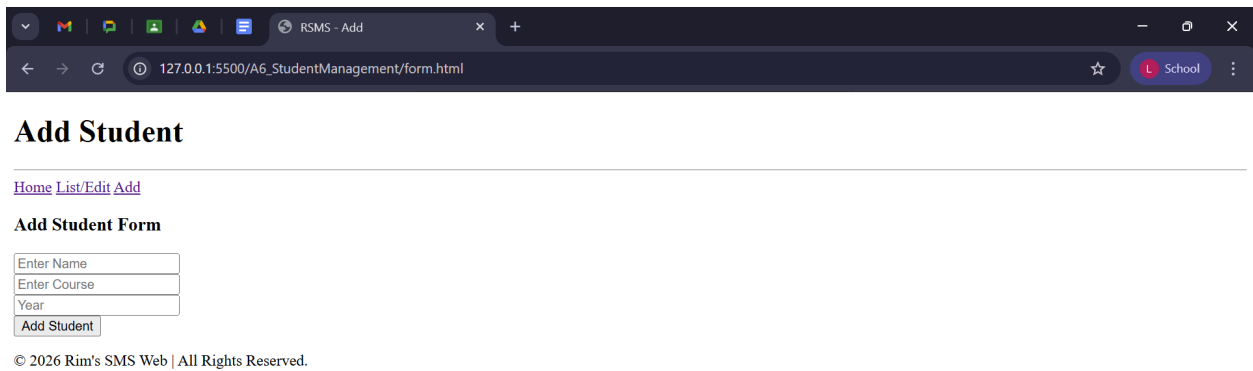
```
<div class="main-content">
  <section class="container">
    <h3>Add Student Form</h3>
    <input type="text" id="name" placeholder="Enter Name"><br>
    <input type="text" id="course" placeholder="Enter Course"><br>
    <input type="number" id="year" placeholder="Year"><br>

    <button onclick="saveStudent()">Add Student</button>

  </section>
</div>
```

I implemented input fields for the user to enter a name, course, and year. Also, another button to save student data.

Screenshot of the web page:



CSS Styling

1. style.css

Here, I started to add designs to style all the pages to create a visually appealing layout.

Whole Code Snippet:

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: Arial, sans-serif;
  background: linear-gradient(to bottom, #33008c, #000000);
  min-height: 100vh;
  display: flex;
  flex-direction: column;
}

header {
```

```
background: #170231;
color: white;
padding: 25px 20px 20px 20px;
text-align: center;
position: relative;
z-index: 10;
}

header h1 {
margin-bottom: 25px;
}

header nav {
margin-top: 20px;
}

nav a {
color: white;
margin: 0 30px;
padding: 5px 10px;
text-decoration: none;
font-weight: bold;
}

nav a:hover {
background: #3498db;
border-radius: 5px;
}

.main-content {
flex: 1;
}

.container {
width: 80%;
max-width: 700px;
margin: 30px auto;
padding: 25px;
background: #d395ff;
border-radius: 10px;
text-align: center;
}

.top-row {
display: flex;
align-items: center;
gap: 20px;
}

.home-text {
margin-top: 15px;
```

```
    text-align: left;
}

button {
    margin: 10px;
    padding: 10px;
    border: none;
    background: #3498db;
    color: white;
    cursor: pointer;
    border-radius: 5px;
}

button:hover {
    background: #2980b9;
}

input {
    padding: 10px;
    margin: 10px;
    width: 80%;
}

#output p {
    display: flex;
    justify-content: space-between;
    align-items: center;
    background: #f4e3ff;
    padding: 3px;
    padding-left: 10px;
    margin: 10px 0;
    border-radius: 6px;
    border-left: 3px solid #3498db;
}

#output p button {
    padding: 5px 10px;
    font-size: 0.9rem;
}

footer {
    background: #0c0417;
    color: rgb(128, 128, 128);
    text-align: center;
    padding: 15px;
}

html, body {
    height: 100%;
    margin: 0;
    padding: 0;
}
```

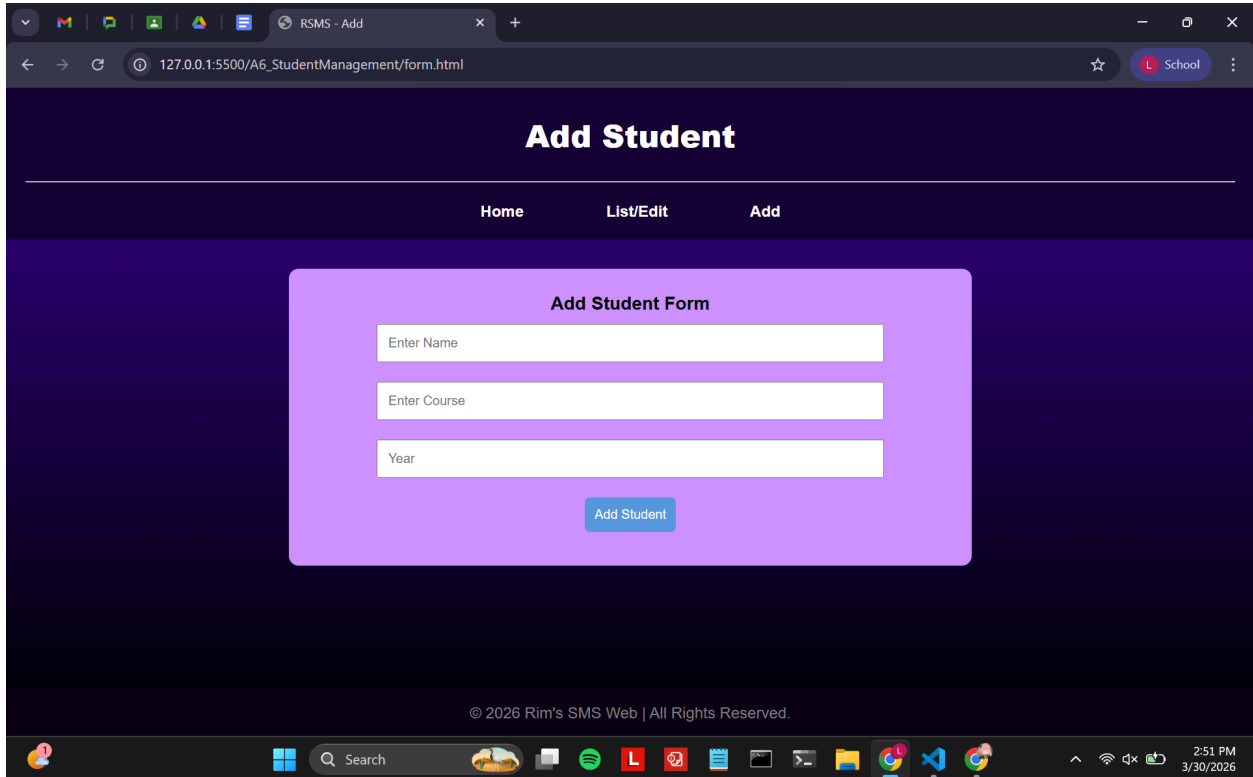
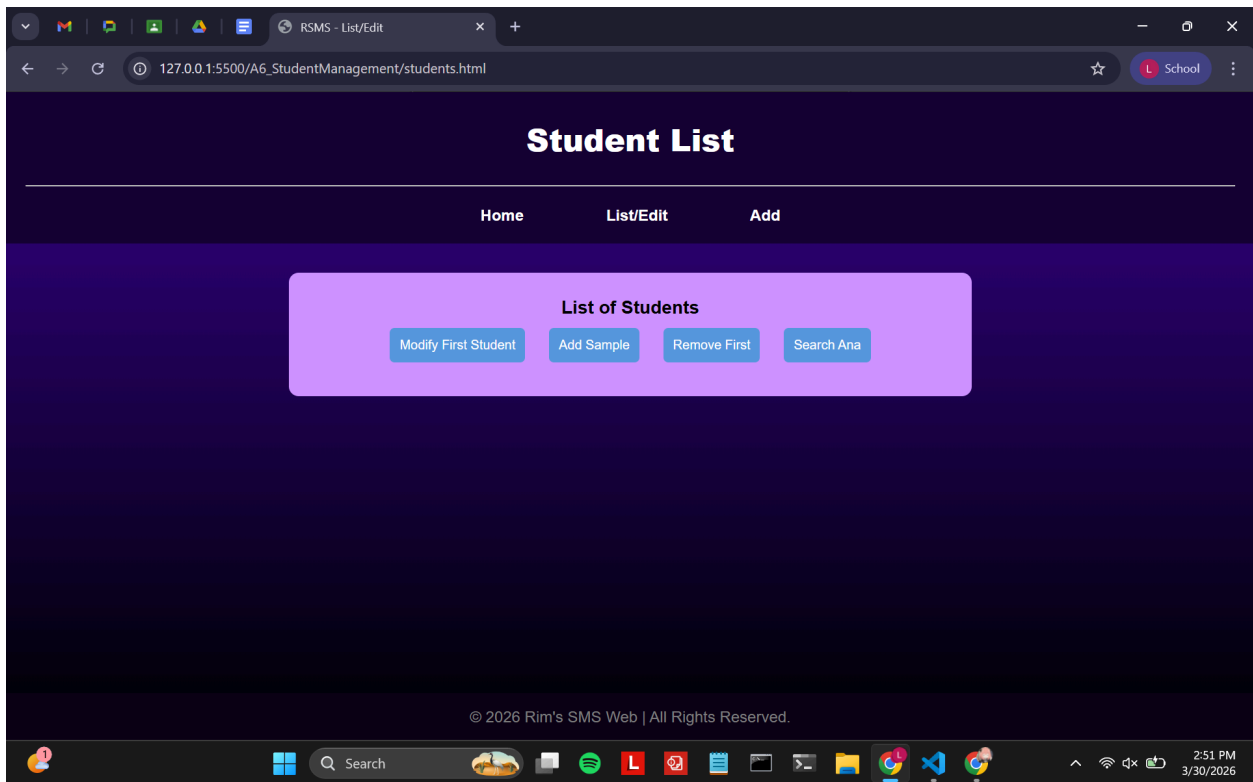
```
}
```

Changes made:

- Gradient background for body appearance
- Centered layout using a container for main content
- Styled navigation bar and buttons with hover effects
- Input fields styled for readability
- Organized layout using spacing, padding, and alignment

Screenshot of the web pages:





JavaScript Implementation

1. script.js

This file will handle all the functionality of the system, including storing, displaying, and modifying student data. The system uses an array of objects to store student data. Each student contains a name, course, and year level.

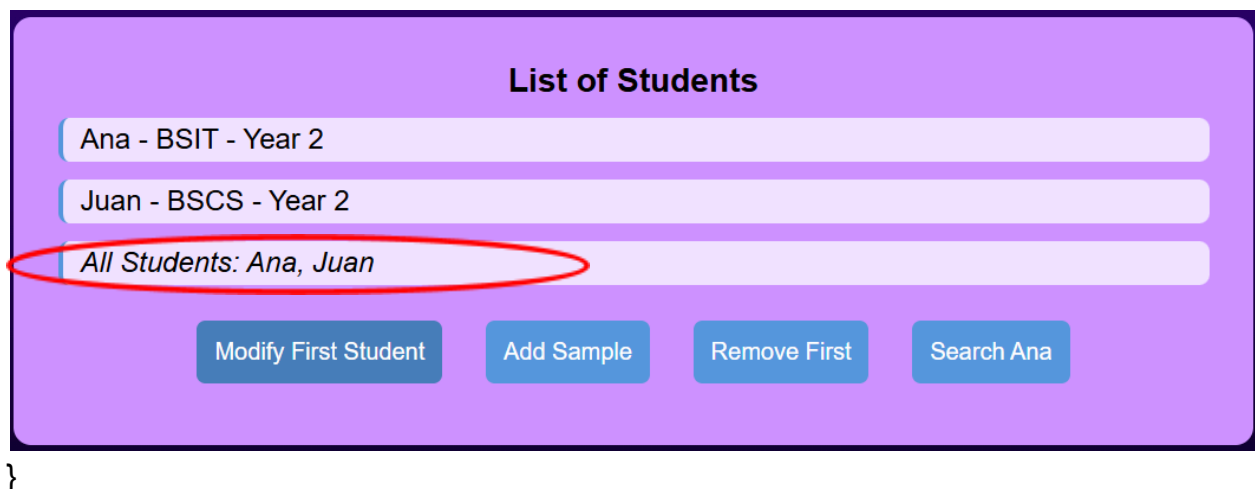
```
{
function displayStudents() {
  let output = document.getElementById("output");
  if (!output) return;

  output.innerHTML = "";

  for (let i = 0; i < students.length; i++) {
    output.innerHTML += `
      <p>
        ${students[i].name} - ${students[i].course} - Year ${students[i].year}
      </p>
    `;
  }

  let names = students.map(s => s.name);
  output.innerHTML += `<p><i>All Students: ${names.join(", ")}</i></p>`;
}
}
```

This function uses DOM manipulation (innerHTML) to show the student list on the webpage. It also uses the join method to display the student names in a single line.



```

{
function modifyStudent() {
  if (students.length === 0) {
    alert("No students to modify!");
    return;
  }

  let updatedStudent = {
    ...students[0],
    year: 3
  };

  students.splice(0, 1, updatedStudent);
  saveStudents();
  displayStudents();
}
}

```

This function updates the first student's year using the spread operator to copy existing data and the splice() method to replace the object in the array.

Ana - BSIT - Year 3

```

}

```

```

{
function addSample() {
  students.push({ name: "Maria", course: "BSIT", year: 2 });
  saveStudents();
  displayStudents();
}
}

```

This function uses push() to add a predefined student to the list.

List of Students

Ana - BSIT - Year 3

Juan - BSCS - Year 2

Maria - BSIT - Year 2

All Students: Ana, Juan, Maria

Modify First Student

Add Sample

Remove First

Search Ana

```

}

```

```

{
function removeFirst() {
  students.shift();
  saveStudents();
  displayStudents();
}
}

```

This function uses the `shift()` method to remove the first student from the list.

List of Students

Juan - BSCS - Year 2

Maria - BSIT - Year 2

All Students: Juan, Maria

Modify First Student Add Sample Remove First Search Ana

```

}

```

```

{

```

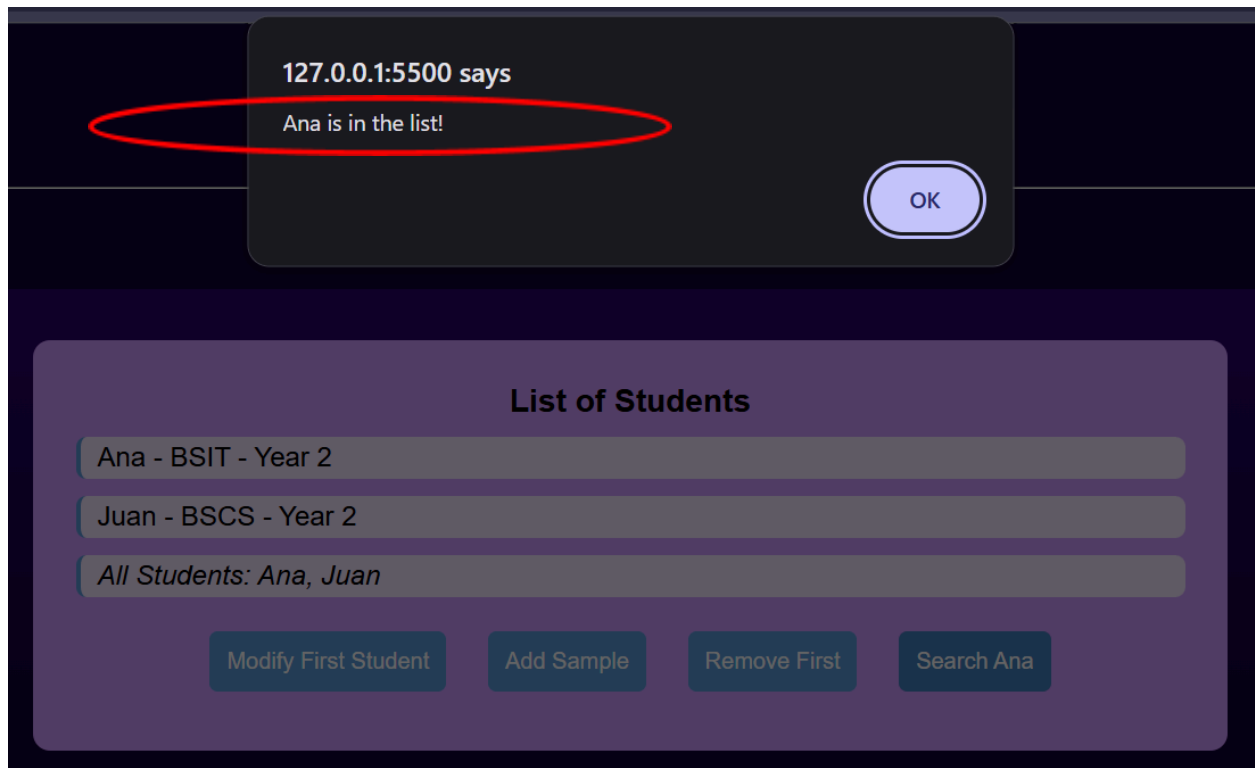
```

function searchStudent() {
  let names = students.map(s => s.name);

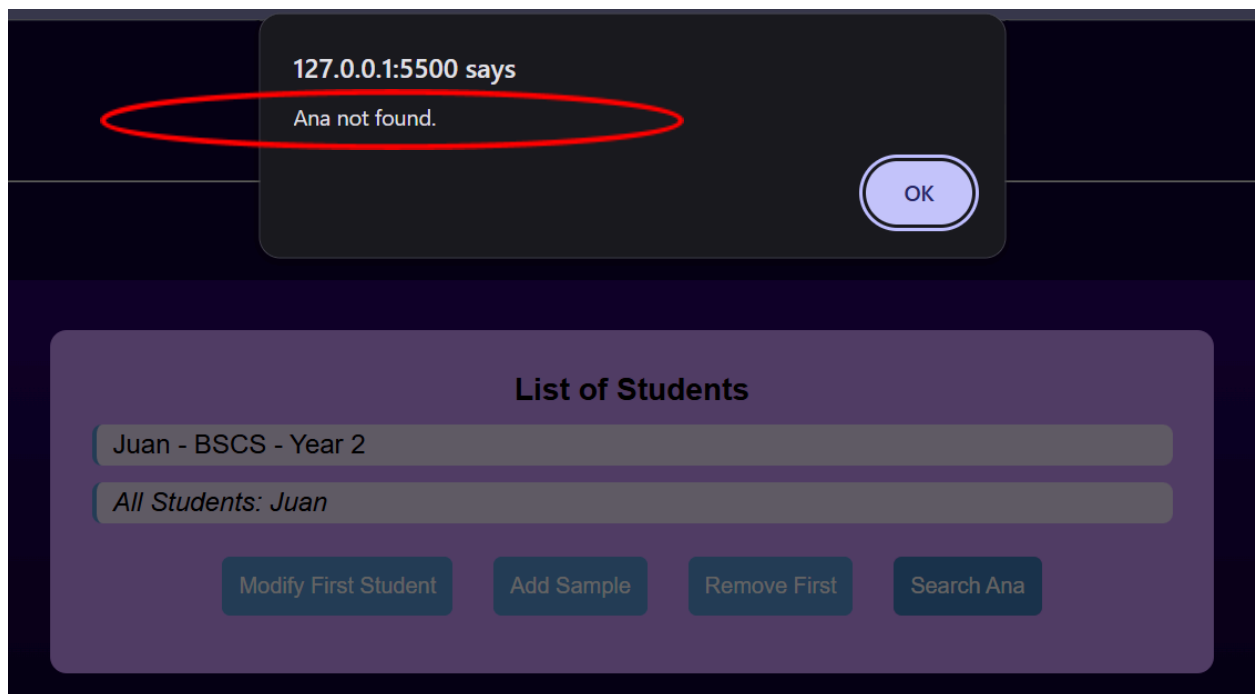
  if (names.includes("Ana")) {
    alert("Ana is in the list!");
  } else {
    alert("Ana not found.");
  }
}
}

```

This function checks if a student exists by name using the `includes()` method.



If there's no 'Ana' in the list, it will alert "Ana not found".



}

```

{
function saveStudent() {
    let name = document.getElementById("name").value;
    let course = document.getElementById("course").value;
    let year = document.getElementById("year").value;

    if (!name || !course || !year) {
        alert("Please fill all fields!");
        return;
    }

    students.push({ name: name, course: course, year: year });

    saveStudents();
    alert("Student added!");
}
}

```

This function allows users to input data from the form and add it to the array. This also uses the `push()` method to insert a new student object.

The screenshot displays a web application interface. At the top, a dark grey alert box with rounded corners contains the text "127.0.0.1:5500 says" in bold, followed by "Student added!" in a lighter font. An "OK" button is located at the bottom right of the alert box. Below the alert, there is a purple rectangular container with rounded corners titled "Add Student Form" in bold. Inside this container, there are three input fields: the first contains "Lupin", the second contains "BSCE", and the third contains "4". At the bottom center of the purple container is a dark blue button with the text "Add Student".

```
{  
  
displayStudents();  

```

This function is called to always show the list of the students.

List of Students

Juan - BSCS - Year 2

Lupin - BSCE - Year 4

All Students: Juan, Lupin

Modify First Student

Add Sample

Remove First

Search Ana